

Numerical Study of Dean Flow Formation in Curved Microchannels

ME 224 – Fundamentals of Microscale Flows

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Abstract

- This study presents a numerical investigation of Dean flow formation in curved microchannels using computational simulations.
- Dean flow occurs when a viscous fluid moves through a curved channel, generating secondary vortices due to centrifugal forces and radial pressure gradients.
- In this work, two microchannel geometries—U-bent and spiral—were modeled and analyzed using ANSYS to study the development and strength of Dean vortices under different flow conditions.
- The simulations were performed by varying the flow velocity, which changed the Reynolds and Dean numbers.
- The results show that as velocity increases, the Dean number also increases, resulting in stronger secondary vortices and a shift of high-velocity regions toward the outer wall of the curved channel.
- These findings demonstrate the role of curvature and flow conditions in enhancing mixing and secondary flow behavior in microfluidic systems, which is important for applications such as biomedical devices, chemical processing, and heat transfer.

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1. Introduction

- Dean flow refers to the fluid motion that occurs when a viscous fluid travels through a curved pipe, tube, or channel, this phenomenon is characterized by the presence of a secondary flow superimposed on the primary axial flow.
- When the pipe curves, the fluid is subjected to a centrifugal force directed outward from the center of curvature. Because the fluid in the core of the pipe is moving at a higher axial velocity than the fluid near the walls, it experiences a stronger centrifugal force.
- This creates a radial pressure gradient. The high-velocity fluid in the center is swept toward the outer wall of the curve. To satisfy the continuity equation, the slower-moving fluid near the outer wall is forced to flow along the pipe perimeter back toward the inner wall. This transverse motion creates a pair of counter-rotating secondary flow cells known as Dean vortices.
- The Dean number (De) is a dimensionless number that describes flow in a curved pipe or channel. It tells us how strong the secondary swirling motion (Dean vortices) will be,

$$De = Re \sqrt{\frac{D}{2R}}$$

Where

- Re = Reynolds number
- D = pipe diameter
- R = radius of curvature

It represents the **ratio of centrifugal forces to viscous forces** in curved flow.

$$\boxed{\text{Vortex strength} \propto De^2}$$

2. Literature Review

- Foundational Theory: W.R. Dean's original mathematical formulations for secondary flow in curved pipes.
- Note on the motion of fluid in a curved pipe. Published online by Cambridge University Press: 26 February 2010.
- Direct comparison of secondary flow development between U-bent and spiral configurations under specific boundary conditions.

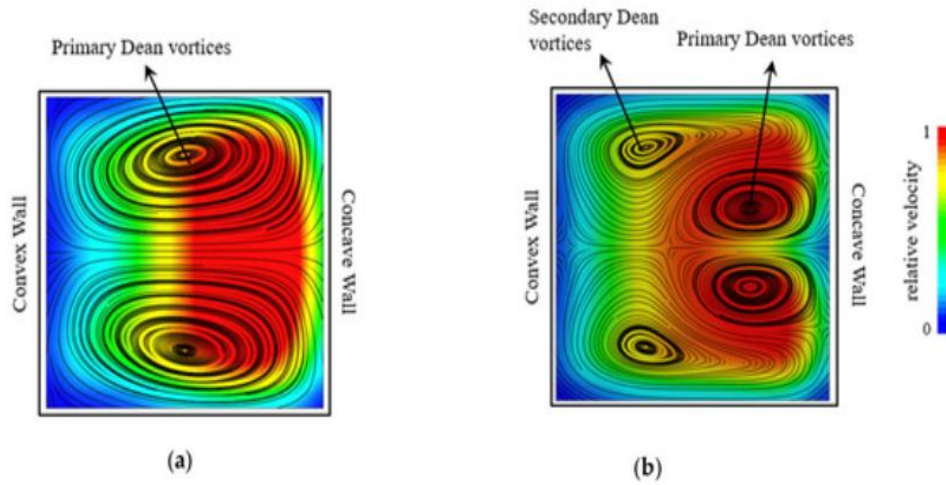


Fig. Rectangular cross section

3. Motivation and Objective

Motivation: The need for highly efficient, passive, and compact microfluidic devices for biological assays, chemical synthesis, or heat transfer.

Primary Objective: To numerically simulate and analyze the formation and evolution of Dean vortices in curved microchannels.

Objective:

1. To develop 3D computational models of a U-bent microchannel and a spiral microchannel using ANSYS.
2. To analyze the Dean vortex strength across varying Reynolds/Dean numbers.

4. Methodology

- Governing Equations: 3D steady, incompressible Continuity and Navier-Stokes equations

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\rho(\mathbf{V} \cdot \nabla) \mathbf{V} = -\nabla p + \mu \nabla^2 \mathbf{V} + \mathbf{F}$$

- Dean Number:

$$De = Re \sqrt{\frac{D_h}{2R_c}}$$

- Reynolds Number:

$$Re = \frac{\rho V D}{\mu}$$

- Computational Tool: ANSYS
- Geometry Designs: U-Bent Microchannel and Spiral Microchannel

Numerical analysis by varying velocity of flow, thus the Reynolds Number changes and hence Dean Number varies giving us a comparison of dean strength.

I. U-Bent Microchannel

Why U-shaped Microchannel?

- Produces constant curvature, leading to a constant Dean number.
- Simplest curved geometry to isolate Dean flow physics.
- Enables clear observation of symmetric counter-rotating Dean vortices.

Dimensions & Reasoning

Channel Cross-Section (200 μm $\times$ 100 μm)

- Chosen to ensure laminar flow in the microfluidic regime.
- Aspect ratio (2:1) enhances clear and stable Dean vortices.
- Rectangular geometry is easy to fabricate and numerically stable

Radius of Curvature (750 μm)

- Selected to generate sufficient centrifugal force for Dean flow.
- Keeps curvature ratio $D/2R < 0.1$, ensuring flow stability.
- Enables Dean vortices at low Reynolds numbers.

Bend Angle (180° U-bend)

- Provides adequate curved length for full development of Dean vortices.
- Eliminates partial or undeveloped secondary flow effects.

Inlet and Outlet Lengths (10D)

- Ensures fully developed flow before entering the bend.
- Minimizes outlet boundary influence on secondary flow structures.

II. Spiral Microchannel

Why Spiral-shaped Microchannel?

- A spiral microchannel provides a continuously varying curvature, leading to a gradual increase in Dean number along the flow direction. This allows systematic investigation of the evolution and strengthening of Dean vortices within a single device. Spiral geometries are widely used in inertial microfluidics due to their ability to generate strong secondary flows over extended lengths.

Dimension & Reasoning

Channel Cross-Section (~1.30 mm Width, Rectangular)

- Selected to maintain laminar flow conditions and a constant hydraulic diameter.
- The rectangular cross-section ensures fabrication simplicity, numerical robustness, and clear visualization of curvature-induced secondary flows.

Radius of Curvature ($R = 5$ mm)

- A fixed curvature radius generates sufficient centrifugal forces to induce stable Dean vortices while avoiding excessive pressure drop.
- This radius lies in a regime where Dean flow effects are pronounced and repeatable.

Bend Angle ($=180^\circ$ Curved Section)

- The semicircular bend provides adequate curved length for full development of Dean vortices.
- It minimizes entrance effects from upstream straight sections and allows controlled analysis of vortex formation and symmetry.

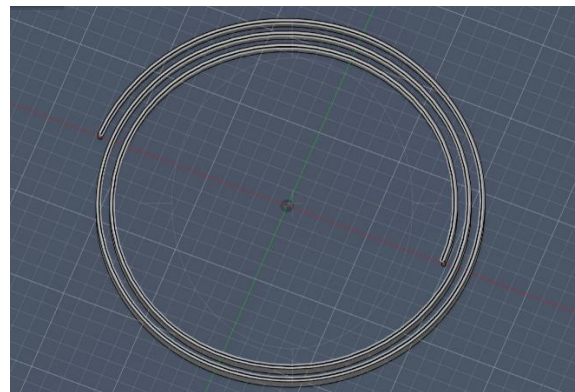
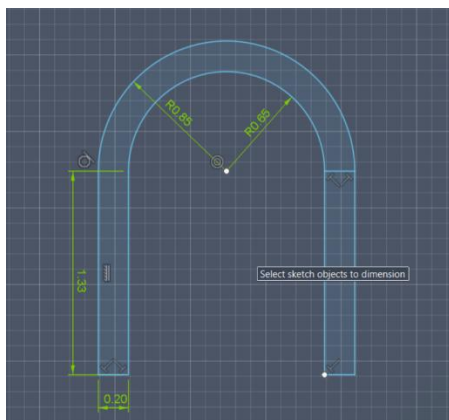


Fig. Schematic diagram of U-Bent Microchannel and Spiral Microchannel

5. Results

I. U-Bent Microchannel

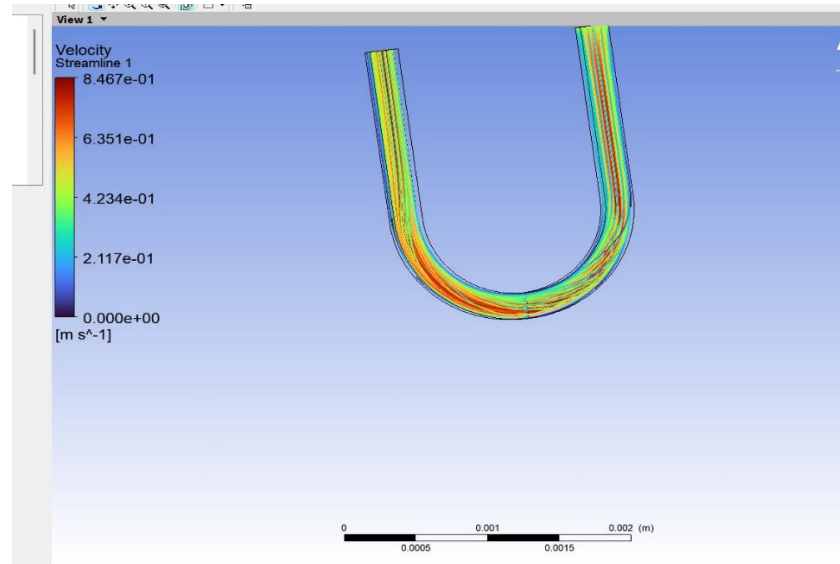


Fig. Simulated U-Bent Microchannel

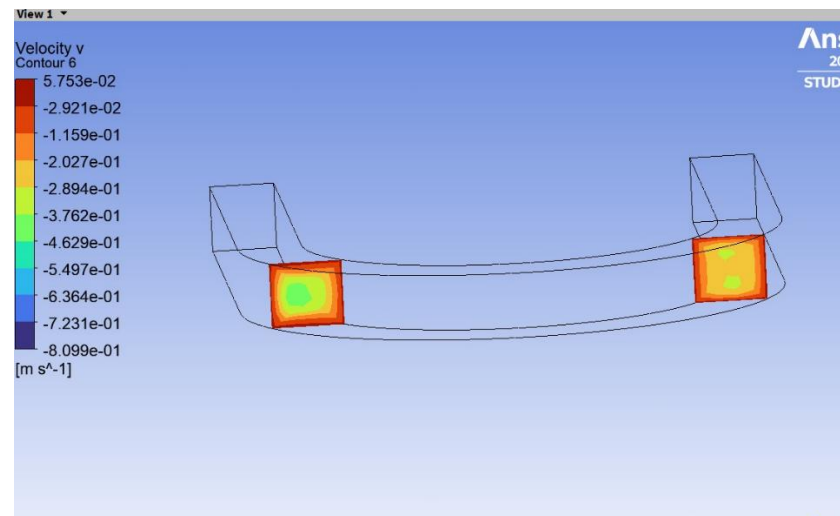
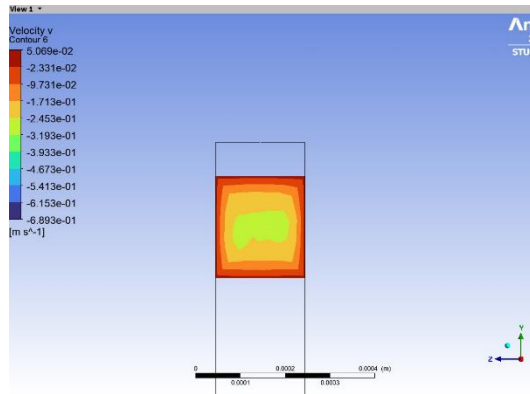
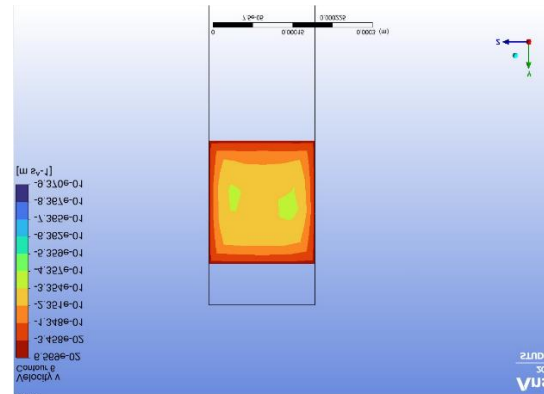


Fig. Dean Vortices

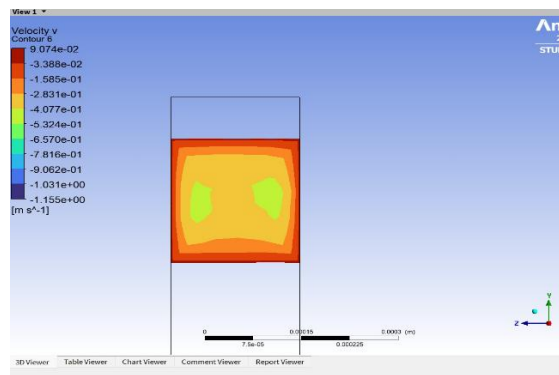
Velocity, v (m/sec)	Reynolds Number, Re	Dean Number, De
0.38	50	14.9
0.53	70	20.9
0.68	90	26.8



(i)



(ii)



(iii)

Fig. Dean vortices at (i) $Re=50$, (ii) $Re=70$, (iii) $Re=90$

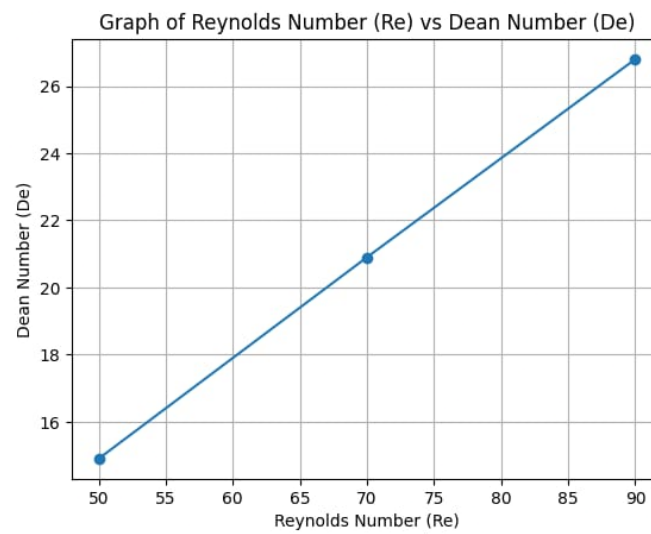


Fig. Linear Graph of Re Vs De

II. Spiral Microchannel

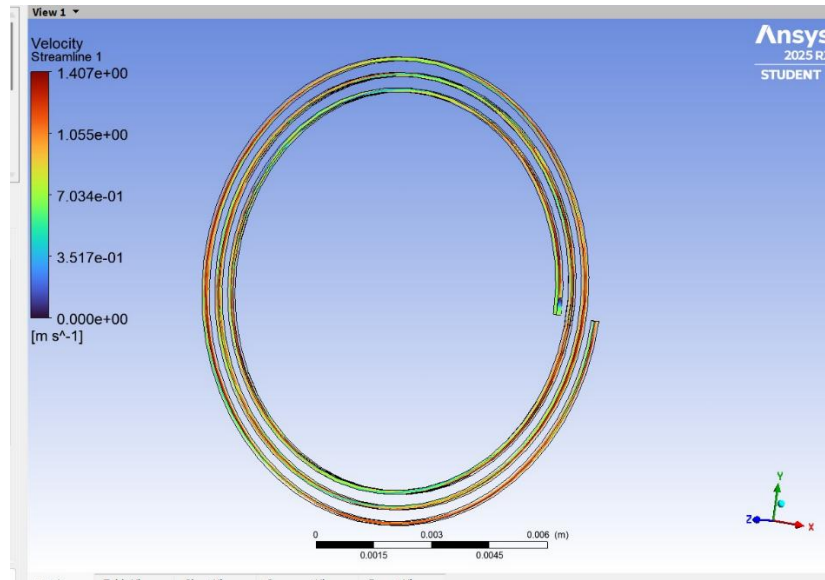


Fig. Simulated Spiral Microchannel

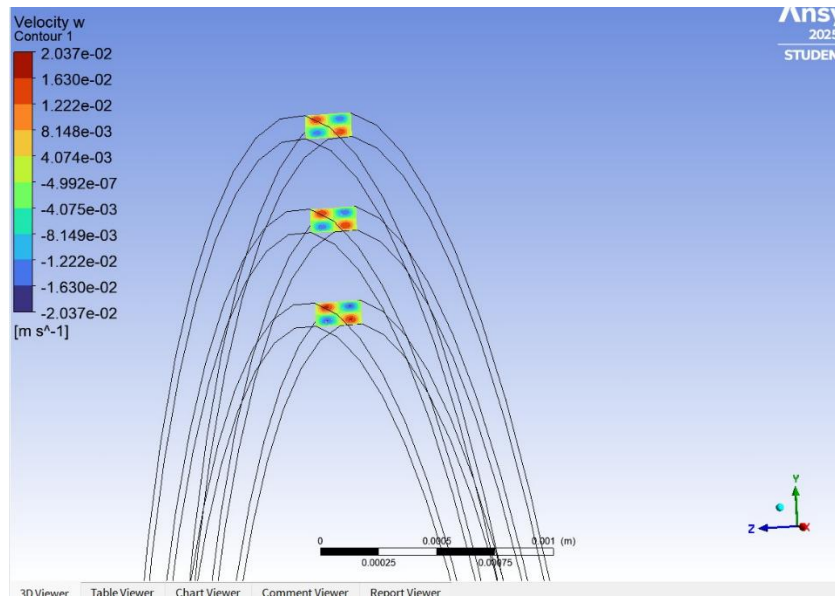
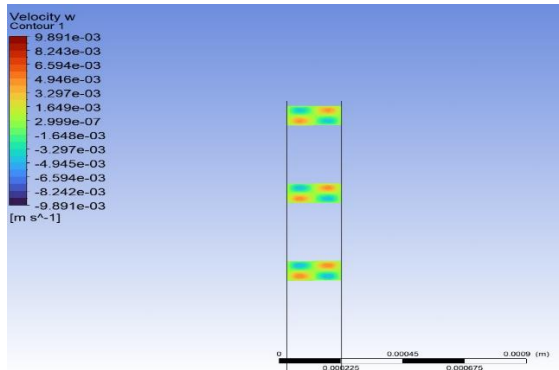
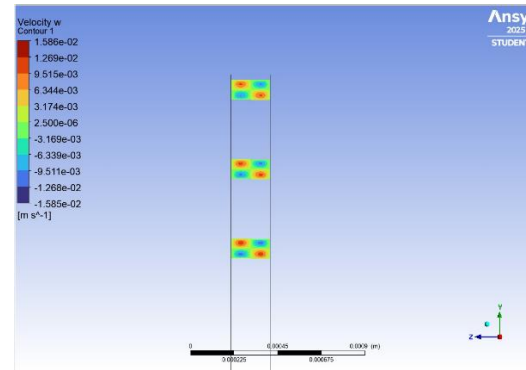


Fig. Dean Vortices

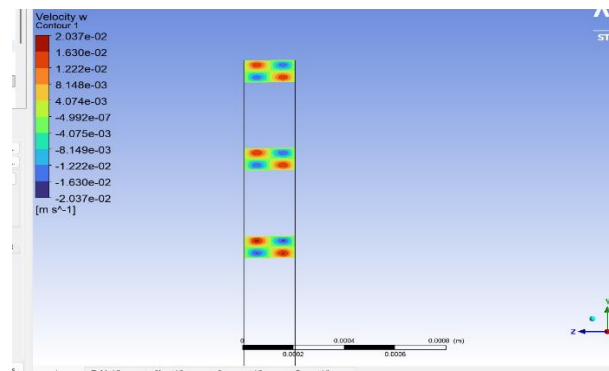
Velocity, v (m/sec)	Reynolds Number, Re	Dean Number, De
0.40	53	6.11
0.60	80	9.17
0.75	100	11.5



(i)



(ii)



(iii)

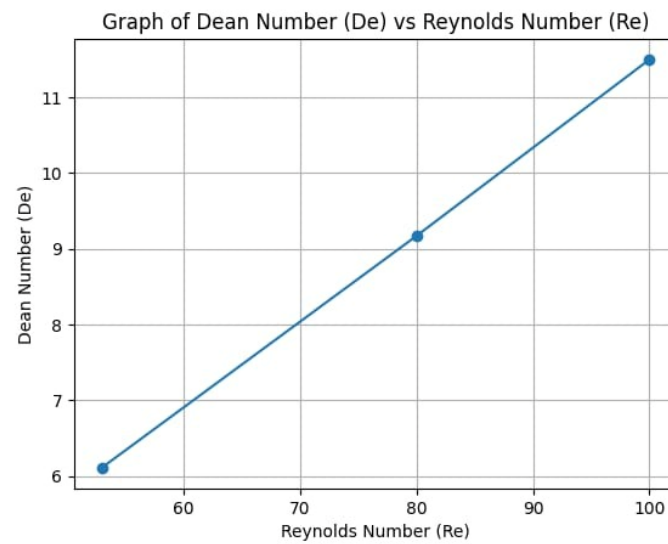
Fig. Dean Vortices at (i) $Re=53$, (ii) $Re=80$, (iii) $Re=100$ 

Fig. Linear Graph of De Vs Re

6. Conclusions

Observation :

- Velocity increases along the bend.
- High velocity region (red) shifts toward the outer wall of the curve.
- Streamlines crowd near the outer wall.
- As velocity increases Dean number increases thus the strength of vortex increases.

Interpretation :

This shift occurs because,

- Centrifugal force pushes high momentum fluid outward.
- A radial pressure gradient balances this force.
- This imbalance generates transverse circulation.
- This is the primary evidence of Dean effect in the simulation.

7. Future Scope

- Heat Transfer Analysis: Enable the energy equation in the simulation setup to study how Dean vortices enhance thermal mixing and temperature distribution within the curved channels.
- Sequential Channel Networks: Connect multiple U-bends in a series (creating a serpentine channel) to observe how the flow develops and whether the vortices amplify or dissipate over a longer network.
- Testing Different Fluids: Modify the fluid properties to simulate non-Newtonian fluids (like blood or polymer solutions) to observe how changing viscosity affects the secondary flow in biomedical applications.

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